

Healthcare Data Privacy and Security

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Strategic Knowledge and Inclusive Lifelong Learning in the health sector for youth Employability and Development

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Learning path

We will move from **what must be protected**, to **where it lives**, to **how it can fail**, and finally to **how security choices reduce risk**.

Hospital storyline

St. Isidore Hospital gives us one continuous scenario: first we identify the systems and security objectives, then we examine risks and attacks, and finally we discuss how design and policy shape safer clinical workflows.

Running Scenario

Section Context: Running Scenario

What we are talking about

Before introducing formal security concepts, we establish a shared hospital environment. This keeps the lesson practical: every definition will be connected to a concrete clinical workflow, system, user, or failure mode.

Why it matters

In healthcare, security decisions are never abstract. A control that looks reasonable on paper can affect triage, medication delivery, lab turnaround time, or patient trust.

The Hospital We Will Use

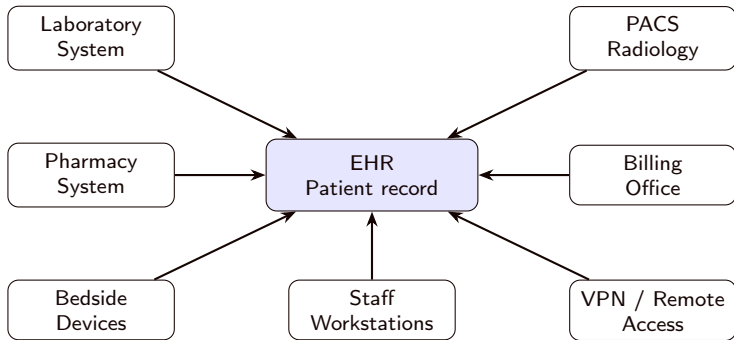
Running example: St. Isidore Hospital

St. Isidore is a mid-size hospital with an emergency department, wards, a laboratory, radiology, pharmacy, billing office, and a small IT team.

Digital systems

The hospital uses an **EHR**, a **laboratory system**, a **PACS** for medical images, a **pharmacy system**, staff laptops, Wi-Fi, remote access for doctors, and a few connected medical devices.

Scheme: Hospital Digital Ecosystem



Reading the scheme

The EHR is central, but security depends on every connected system, user, and network path.

Why This Scenario Matters

Key idea

Security in healthcare is not only about protecting computers. It protects **patients, clinical decisions, care continuity, and trust.**

Hospital example

If the EHR is unavailable during a night shift, clinicians may not see allergies, medications, or recent lab results. A technical failure becomes a *patient-safety* issue.

Recap: Running Scenario

Takeaways

- We will use **St. Isidore Hospital** as the running example throughout the lesson.
- The hospital combines clinical systems, administrative systems, networks, devices, and people.
- Security protects *care delivery*, not only servers and databases.

Security Objectives

Section Context: Security Objectives

What we are talking about

This section explains the goals security is trying to achieve. The key question is: *what does it mean for a hospital system to be secure?*

Hospital lens

For St. Isidore, a secure EHR is not merely hidden from outsiders. It must be readable by the right clinicians, accurate enough to support care, available during work, and traceable when used.

Computer Security

Definition

Computer security is the protection provided by an information system to preserve the security objectives of the system and its resources. NIST frames the core objectives as **confidentiality**, **integrity**, and **availability** [National Institute of Standards and Technology, 2004].

Hospital example

St. Isidore must protect patient records, clinical software, network links, staff accounts, medical images, backup systems, and devices used at the bedside.

System Resources

Concept

A system resource is anything the organisation depends on to deliver information services.

- **Hardware:** servers, laptops, scanners, network devices.
- **Software:** EHR, PACS, pharmacy and laboratory systems.
- **Data:** clinical notes, lab results, images, prescriptions.
- **Telecommunications:** Wi-Fi, VPN, internal network, Internet links.

Hospital example

An MRI image is data; the PACS is software; the storage server is hardware; the link between radiology and the ward is telecommunications.

The CIA Triad

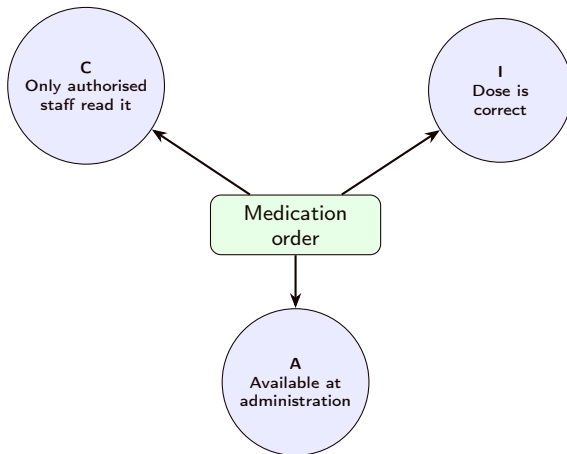
Core model

The CIA triad summarises three central security goals: **confidentiality**, **integrity**, and **availability**.

Hospital example

For a medication order, **confidentiality** means only authorised staff can see it; **integrity** means the dose is not changed incorrectly; **availability** means the order is reachable when the nurse administers treatment.

Scheme: CIA for a Medication Order



Takeaway: the same clinical object must satisfy all three goals at once.

Confidentiality

Concept

Confidentiality means that information is accessed and disclosed only according to authorised rules.

- It is not secrecy from everyone.
- It is *controlled disclosure* to the right people for the right purpose.

Hospital example

An emergency doctor may view a patient's allergies. A billing clerk may view insurance data. Neither should automatically see psychotherapy notes unrelated to their role.

Privacy

Concept

Privacy concerns how personal information is collected, stored, used, shared, and retained.

- Confidentiality protects access to information.
- Privacy also asks whether the information should be processed at all.

Hospital example

A research team may need aggregated diabetes statistics. Privacy requires minimisation, consent or another lawful basis, and removal of unnecessary identifiers.

Data Integrity

Concept

Data integrity means that information is accurate, complete, and modified only through authorised and expected processes.

Hospital example

If a patient's allergy to penicillin is deleted or changed, the record may still be confidential and available, but it is no longer safe to trust.

System Integrity

Concept

System integrity means that the system performs its intended functions without unauthorised manipulation.

Hospital example

The pharmacy system should calculate interactions and dosage warnings according to approved rules. If malware changes those rules, the system may give unsafe recommendations.

Availability

Concept

Availability means that information and services are reachable when they are needed.

- Outages matter.
- Severe slowness also matters.
- Missing dependencies can break a service even if the application itself is healthy.

Hospital example

If the EHR login service fails, clinicians cannot access records even if the EHR database is still intact.

Authenticity

Concept

Authenticity gives confidence that a user, device, message, or data source is genuine.

Hospital example

When the lab system sends a potassium result to the EHR, the EHR must know it really came from the laboratory system, not from a forged message.

Accountability

Concept

Accountability links actions to identifiable entities. It supports audit, investigation, deterrence, and legal responsibility.

Hospital example

If a celebrity patient is admitted, audit logs should show which staff members opened the record, when, and from which workstation.

Recap: Security Objectives

Takeaways

- **Confidentiality** controls who can see sensitive information.
- **Integrity** keeps data and system behaviour trustworthy.
- **Availability** keeps clinical services reachable when needed.
- **Authenticity** and **accountability** make actions reliable and traceable.

Hospital checkpoint

A medication order is secure only if it is seen by authorised staff, unchanged, available at administration time, issued by a genuine clinician, and auditable afterwards.

Components and Networks

Section Context: Components and Networks

What we are talking about

This section looks at where threats land in a real system: hardware, software, data, and communication lines. Network security matters because hospital systems are connected by many paths and dependencies.

Hospital lens

An EHR outage may start with a stolen laptop, a vulnerable application, corrupted data, or a network attack. Understanding the component helps choose the right control.

Hardware Threats

Concept

Hardware is especially exposed to physical threats: theft, damage, tampering, and loss.

Hospital example

A laptop containing cached patient data is stolen from a clinic room. Full-disk encryption and remote wipe reduce confidentiality risk.

Software Threats

Concept

Software can be deleted, copied, misconfigured, or modified so that it behaves differently from what users expect.

Hospital example

A malicious browser extension on a workstation captures EHR session data. Application allowlisting and endpoint monitoring help reduce this risk.

Data Threats

Concept

Data must be protected for confidentiality, integrity, and availability at the same time.

Hospital example

Backups protect availability, access control protects confidentiality, and audit trails help detect unauthorised modification of patient records.

Communication-Line Threats

Concept

Network communication can be read, blocked, delayed, modified, duplicated, or reordered.

Hospital example

A lab result message delayed between the laboratory system and the EHR may cause clinicians to act on outdated information.

Passive Network Attacks

Concept

Passive network attacks observe communication without changing it. They are hard to detect because systems continue to work normally.

Hospital example

Traffic analysis shows that a VIP patient's room communicates frequently with oncology and radiology, revealing sensitive context without reading message contents.

Active Network Attacks

Concept

Active network attacks modify traffic, create false traffic, or prevent normal communication.

Hospital example

An attacker performs a replay attack by resending an old authentication message from a medical device. Fresh nonces, timestamps, and mutual authentication reduce this risk.

Denial of Service

Concept

Denial of service prevents legitimate users from accessing a channel, system, or service.

Hospital example

If the medication-ordering module becomes unavailable, nurses may switch to paper procedures. That fallback must be planned, trained, and reconciled later.

Recap: Components and Networks

Takeaways

- Hardware, software, data, and networks fail in different ways.
- Network traffic exposes both **content** and **metadata**.
- Passive attacks are hard to notice because they do not alter traffic.
- Active attacks require prevention, detection, response, and fallback procedures.

Hospital checkpoint

The lab-to-EHR connection must protect confidentiality, integrity, and availability because delayed or altered results can change clinical decisions.

Impact and Risk

Section Context: Impact and Risk

What we are talking about

This section connects security problems to consequences. We move from *what could happen* to *how bad it would be* and *how we decide what to prioritise*.

Hospital lens

St. Isidore cannot protect every asset with the same intensity. The visitor parking page and the medication-ordering module need different levels of protection because their failures have different clinical impact.

Impact Levels

Concept

Impact describes how harmful a security incident would be. NIST FIPS 199 uses **low**, **moderate**, and **high** impact levels [National Institute of Standards and Technology, 2004].

Hospital example

The public cafeteria menu has low security impact. The EHR medication module has high impact because failure may affect patient treatment.

Low Impact

Low impact

The incident causes limited adverse effects. The organisation can still perform its primary functions, perhaps with reduced efficiency.

Hospital example

The hospital public website loses the page listing visitor parking instructions. This is annoying, but it does not stop care delivery.

Moderate Impact

Moderate impact

The incident causes serious adverse effects, such as significant disruption, financial loss, or non-life-threatening harm.

Hospital example

The appointment booking system is down for a day. Clinics continue, but staff must reschedule patients manually and delays accumulate.

High Impact

High impact

The incident causes severe or catastrophic effects, including long-term mission failure, major loss, severe injury, or death.

Hospital example

Ransomware disables the EHR, laboratory results, and pharmacy ordering during emergency care. The hospital must switch to paper while treating critical patients.

Impact Depends on the Security Goal

Concept

The same asset can have different impact levels for confidentiality, integrity, and availability.

Hospital example

A public ward phone number has low confidentiality impact, moderate availability impact during family coordination, and low integrity impact unless it redirects patients to the wrong service.

Why Security Is Difficult

Concept

Security requirements sound simple, but the mechanisms interact with technology, workflow, people, physical space, and law.

Hospital example

Multi-factor authentication protects remote EHR access. But if it blocks surgeons during an emergency, the hospital needs an emergency-access procedure that is fast, logged, and reviewed.

Security Is Asymmetric

Concept

An attacker may need one exploitable weakness. Defenders must reduce many weaknesses and keep doing so as systems change.

Hospital example

The IT team patches servers, trains staff, segments networks, reviews logs, and tests backups. An attacker may only need one unpatched VPN appliance or one successful phishing email.

Vocabulary

The Internet security glossary defines key terms used across security practice [Shirey, 2007].

- **Asset:** something valuable to protect.
- **Threat:** a possible cause of harm.
- **Vulnerability:** a weakness that can be exploited.
- **Attack:** a deliberate attempt to violate security.
- **Countermeasure:** a control that reduces risk.

Core Terms in the Hospital

Hospital example

Asset: EHR database.

Threat: ransomware group.

Vulnerability: exposed remote desktop service.

Attack: brute-force login followed by malware deployment.

Countermeasure: VPN, MFA, account lockout, monitoring, backups.

Takeaway

Precise language helps teams discuss incidents without mixing up causes, weaknesses, actions, and consequences.

Risk



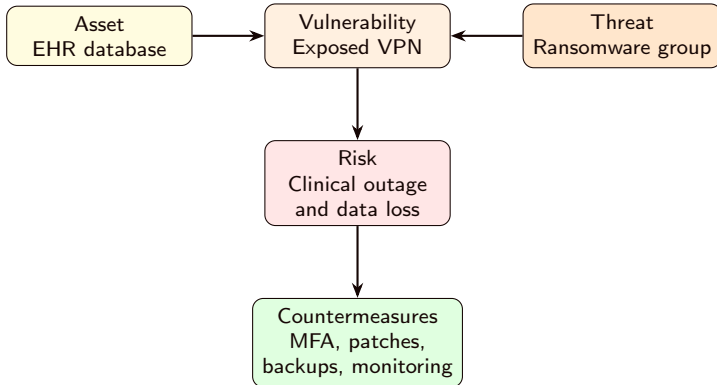
Concept

Risk is the expected loss when a threat exploits a vulnerability and causes impact. NIST SP 800-30 relates risk to threat sources, vulnerabilities, likelihood, and impact [Initiative, 2012].

Hospital example

An exposed VPN service is risky when attackers are actively exploiting that product and the VPN allows access to clinical systems.

Scheme: From Weakness to Risk



Reading the scheme

Risk emerges when a valuable hospital asset is reachable through an exploitable weakness.

Risk Matrix

Concept

A risk matrix compares **likelihood** and **impact**. It is a communication tool, not a mathematical proof.

Hospital example

A rare failure in a public display screen may be low priority. A likely phishing campaign against nurses with access to medication orders deserves immediate attention.

Estimating Likelihood

Concept

Likelihood is an informed judgement about how plausible an event is in a defined time period. It should combine evidence about the threat, the weakness, and the hospital environment.

- **Threat activity:** are attackers currently exploiting this kind of target?
- **Exposure:** is the vulnerable system reachable from the Internet, VPN, Wi-Fi, or vendors?
- **Control strength:** are MFA, patches, monitoring, and segmentation effective?
- **History:** has the hospital or a peer organisation seen similar incidents?

Likelihood in the Hospital

Hospital example

For St. Isidore, phishing against staff is **high likelihood**: it is common, cheap, and targets busy users. Theft of a server from the data centre is **low likelihood**: physical access is restricted and monitored.

Practical rule

Do not estimate likelihood from intuition alone. Use logs, vulnerability scans, incident history, threat intelligence, audit findings, and knowledge of clinical workflows.

Example Risk Matrix

	Low impact	Moderate impact	High impact
High likelihood	Phishing email to generic mailbox	Phishing nurses with portal access	Exploited reaching systems VPN clinical
Medium likelihood	Public screen misconfiguration	Appointment system outage	Malware on ward workstation
Low likelihood	Cafeteria page defacement	Theft of encrypted laptop	Data-centre fire with tested failover

Reading the matrix

Items in red or orange are discussed first. Green items are not ignored, but they usually receive lighter controls or later treatment.

Residual Risk

Concept

Residual risk is the risk that remains after countermeasures are applied.

Hospital example

MFA reduces account takeover, but a tired doctor may still approve a fraudulent push request. Training, number matching, monitoring, and help-desk procedures reduce the remaining risk.

Recap: Impact and Risk

Takeaways

- Impact levels help separate minor inconvenience from patient-safety risk.
- Risk combines **asset value**, **threats**, **vulnerabilities**, likelihood, and impact.
- A risk matrix helps prioritise, but it does not remove uncertainty.
- Controls reduce risk; they rarely remove it completely.

Hospital checkpoint

The same ransomware threat is more urgent when it can reach the EHR, lab, and pharmacy systems than when it can only affect an isolated public kiosk.

Vulnerabilities and Attacks

Section Context: Vulnerabilities and Attacks

What we are talking about

This section explains how security failures start. A vulnerability is the weakness; an attack is the attempt to exploit it; the attacker may be inside or outside the organisation.

Hospital lens

At St. Isidore, the same patient record can be threatened by a stolen password, a curious employee, an exposed server, or a malware-infected workstation. The defence depends on the attack path.

Vulnerability Categories

Concept

Vulnerabilities can be grouped by the security property they endanger.

- **Corrupted:** integrity is lost.
- **Leaky:** confidentiality is lost.
- **Unavailable:** availability is lost.

Hospital example

A wrong lab value is *corrupted*; a public folder with patient PDFs is *leaky*; a locked ransomware database is *unavailable*.

Active Attacks

Concept

An **active attack** alters resources or influences system behaviour.

Hospital example

An attacker changes a discharge summary, modifies DNS settings, installs malware on a nurse workstation, or floods a patient portal until it stops responding.

Passive Attacks

Concept

A **passive attack** observes or collects information without directly changing the system.

Hospital example

Someone connected to an insecure guest Wi-Fi captures unencrypted traffic and learns which devices communicate with the radiology system.

Insider Attacks

Concept

An **insider attack** is performed by someone who already has authorised access but uses it in an unauthorised way.

Hospital example

A staff member opens the record of a neighbour without a care-related reason. The account is valid; the use is not.

Outsider Attacks

Concept

An **outsider attack** comes from outside the security perimeter: criminals, opportunistic attackers, hostile actors, or compromised third parties.

Hospital example

A criminal group sends phishing emails to hospital staff to steal credentials and access the EHR remotely.

Preventive Countermeasures

Concept

Preventive controls try to stop an attack before it succeeds.

Hospital example

St. Isidore uses MFA for remote access, network segmentation between guest Wi-Fi and clinical systems, endpoint protection, patched servers, and role-based access control.

Detect and Recover

Concept

Detect-and-recover controls assume that prevention can fail. They identify attacks, contain damage, and restore trustworthy service.

Hospital example

The security team monitors unusual EHR access, isolates infected laptops, restores clean backups, and reviews logs after an incident.

Recap: Vulnerabilities and Attacks

Takeaways

- Vulnerabilities can make systems **corrupted**, **leaky**, or **unavailable**.
- Active attacks change systems; passive attacks observe them.
- Insider attacks misuse legitimate access; outsider attacks cross the perimeter.
- Prevention matters, but detection and recovery are essential when prevention fails.

Hospital checkpoint

An exposed VPN, a stolen account, and weak backup protection are different weaknesses, but together they can form one ransomware path.

Threat Classes

Section Context: Threat Classes

What we are talking about

This section classifies common threat patterns. Instead of memorising isolated attack names, we connect each threat to the security property it damages: confidentiality, integrity, availability, or control of resources.

Hospital lens

The hospital scenario helps distinguish *leaking a record*, *changing a record*, *blocking a service*, and *taking over a system*. Each failure calls for different controls and response procedures.

Unauthorised Disclosure

Concept

Unauthorised disclosure is a confidentiality threat: information reaches an entity that should not receive it.

Hospital example

A misconfigured cloud storage bucket exposes discharge letters containing patient names, diagnoses, and medication lists.

Exposure

Concept

Exposure can be intentional or accidental. It occurs when sensitive information is made visible to the wrong audience.

Hospital example

A spreadsheet of HIV clinic appointments is emailed to a public mailing list by mistake.

Interception

Concept

Interception occurs when communication is observed in transit.

Hospital example

An attacker captures unencrypted traffic between an old bedside device and a monitoring server.

Inference



Concept

Inference means deriving sensitive information from indirect signals, partial data, or metadata.

Hospital example

Even if message contents are encrypted, repeated traffic from oncology to a patient's room at specific times may reveal sensitive clinical context.

Intrusion

Concept

Intrusion is unauthorised access that bypasses controls and reaches protected resources.

Hospital example

An attacker exploits a vulnerable patient portal and downloads appointment histories from the backend database.

Masquerade

Concept

Masquerade is impersonation: one entity pretends to be another.

Hospital example

A stolen nurse account is used to open patient records. To the EHR, the request looks like it came from the nurse unless stronger authentication and anomaly detection intervene.

Falsification

Concept

Falsification replaces valid information with false information.

Hospital example

Changing a blood type, medication dose, or lab result can directly influence clinical decisions. Integrity failures can therefore become safety incidents.

Repudiation

Concept

Repudiation occurs when someone denies an action, such as sending, approving, or receiving data.

Hospital example

A physician denies approving a high-risk medication order. Strong authentication, audit logs, and timestamps help reconstruct what happened.

Disruption

Concept

Disruption threatens availability by preventing normal service.

Hospital example

A denial-of-service attack makes the patient portal unreachable during vaccination booking. Care continues, but administrative pressure rises quickly.

Incapacitation

Concept

Incapacitation disables a system or component through physical damage, malware, or severe failure.

Hospital example

Ransomware encrypts the file server used by radiology. Images cannot be retrieved until recovery procedures start.

Obstruction

Concept

Obstruction interferes with communication or overloads resources.

Hospital example

An attacker floods the VPN gateway. Doctors working from home cannot connect to review urgent test results.

Usurpation

Concept

Usurpation occurs when an unauthorised entity takes control of a logical or physical resource.

Hospital example

Compromised hospital workstations become part of a botnet and are used to attack another organisation, damaging both security and reputation.

Recap: Threat Classes

Takeaways

- **Disclosure** attacks expose information through exposure, interception, inference, or intrusion.
- **Deception** attacks undermine trust through masquerade, falsification, or repudiation.
- **Disruption** attacks reduce availability through incapacity or obstruction.
- **Usurpation** means unauthorised control of hospital resources.

Hospital checkpoint

If a forged lab message is accepted by the EHR, the problem is not only technical: it can affect diagnosis, treatment, auditability, and patient safety.

Attack Surface

Section Context: Attack Surface

What we are talking about

This section brings the lesson together by asking where an attacker can touch the hospital environment. Attack surface analysis maps entry points, weaknesses, paths, controls, and assets.

Hospital lens

For St. Isidore, the attack surface includes technology and people: VPN, email, patient portal, medical devices, vendor access, APIs, Wi-Fi, cloud systems, and staff workflows.

Attack Surface

Concept

The **attack surface** is the set of points where an attacker can interact with, influence, or observe a system.

Hospital example

The patient portal, VPN, Wi-Fi, email, medical devices, vendor support accounts, APIs, and staff workstations are all part of the hospital attack surface.

Network Attack Surface

Concept

The network attack surface includes exposed ports, services, protocols, remote access paths, and network boundaries.

Hospital example

An old remote administration service left open on a radiology server creates a direct path into a clinical network segment.

Software Attack Surface

Concept

The software attack surface includes input channels, APIs, file uploads, web forms, parsers, legacy components, and configuration interfaces.

Hospital example

A patient portal upload feature accepts PDF documents. If parsing is vulnerable, a malicious file may compromise the server.

Human Attack Surface

Concept

The human attack surface includes phishing, social engineering, unclear procedures, fatigue, and pressure during urgent work.

Hospital example

A caller pretending to be from IT asks a nurse to approve an MFA prompt. Training and clear help-desk procedures reduce this path.

Analysing Attack Paths

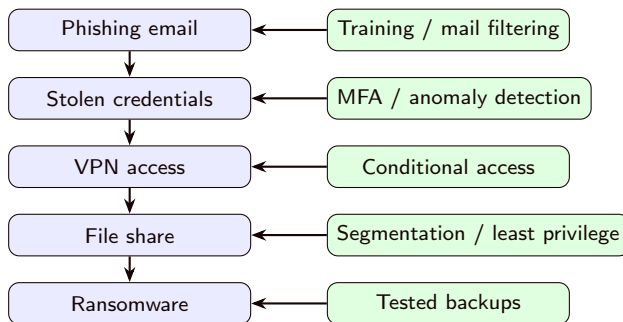
Concept

Attack surface analysis maps entry points, vulnerabilities, controls, and assets that could be reached.

Hospital example

Phishing email → stolen credentials → VPN access → file share → ransomware. Controls can interrupt the path at several points.

Scheme: Example Attack Path



Takeaway

Controls are stronger when they interrupt the path before attackers reach clinical assets.

Reducing Attack Surface

Concept

Reducing attack surface means removing unnecessary exposure and strengthening unavoidable exposure.

Hospital example

St. Isidore closes unused ports, removes stale accounts, segments medical devices, limits vendor access windows, and monitors remote access.

Recap: Attack Surface

Takeaways

- Attack surface is the set of places an attacker can interact with or observe a system.
- Network, software, and human surfaces often connect into one attack path.
- Reducing exposure is as important as strengthening unavoidable exposure.
- Attack-path thinking helps place controls before attackers reach clinical assets.

Hospital checkpoint

Phishing → stolen credentials → VPN → file share → ransomware is a path. MFA, segmentation, least privilege, monitoring, and backups interrupt it at different points.

Design and Policy

Section Context: Design and Policy

What we are talking about

This section shifts from attacks to design choices. Secure systems are not created only by adding tools; they are shaped by principles, policies, workflows, and evidence that controls actually work.

Hospital lens

St. Isidore must design access rules, emergency override, remote access, logging, backup, and incident response so that security supports care instead of fighting it.

Secure Design Principles

Concept

Classic secure design principles reduce complexity, minimise trust, and make failures safer [Saltzer and Schroeder, 1975].

Hospital example

St. Isidore applies these principles when designing EHR access, emergency override, remote access, network segmentation, audit logging, and backup recovery.

Economy of Mechanism

Principle

Security mechanisms should be as simple as possible, so they can be understood, implemented, and verified.

Hospital example

A simple role model for nurse, physician, pharmacist, and administrator is easier to review than hundreds of undocumented special permissions.

Fail-Safe Defaults

Principle

Access should be denied unless permission is explicitly granted.

Hospital example

New temporary staff accounts should start with no EHR access. Access is added only after role, department, and supervisor approval are confirmed.

Complete Mediation

Principle

Every access should be checked, not only the first one.

Hospital example

If a doctor changes department, the EHR should re-check access rights on each record request, not rely on permissions cached months earlier.

Principle

Security should not depend on keeping the mechanism secret. Secrets should be limited to keys, passwords, and credentials.

Hospital example

The hospital should use well-reviewed encryption protocols rather than a custom “secret” algorithm for transmitting lab results.

Least Privilege

Principle

Users, services, and devices should receive only the permissions needed for their tasks.

Hospital example

A kiosk used for patient check-in should not have permission to browse the full EHR database. If compromised, its limited role reduces damage.

Separation of Privilege

Principle

Critical actions should require more than one condition, role, or approval.

Hospital example

Changing a patient's identity record may require both administrative approval and a separate verification by the medical records office.

Layering and Defence in Depth

Principle

Layering means using multiple overlapping controls, so one failure does not expose the whole system.

Hospital example

Remote access uses VPN, MFA, device compliance checks, network segmentation, EHR role checks, and logging. Each layer limits what happens if another layer fails.

Scheme: Defence in Depth

Training, monitoring, and incident response

Network segmentation and firewall rules

VPN, MFA, and conditional access

Managed device and endpoint protection

Role-based access and break-glass review

Clinical asset: EHR medication module

Reading the scheme

No single layer is trusted to be perfect. Each layer reduces what an attacker can do next.

Psychological Acceptability

Principle

Security controls should be understandable and usable enough that people can follow them during real work.

Hospital example

If login takes too long during ward rounds, staff may share accounts. A better design protects security while respecting clinical pace.

Security Policy

Concept

A security policy turns security goals into organisational rules: who may access what, under which conditions, with what logging and responsibilities.

Hospital example

The policy states that staff may access records only for care, operations, payment, or approved research. Curiosity access is prohibited even if the account technically works.

Security Trade-Offs

Concept

Security must balance protection, usability, cost, performance, legal duties, and patient care.

Hospital example

Strict session timeouts protect unattended workstations, but overly aggressive timeouts may delay care. The hospital may use badge tap-in, short idle locks, and fast re-authentication.

Security Lifecycle Overview

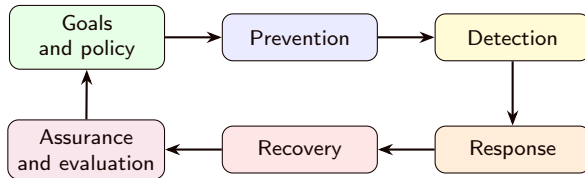
Concept

Security is not a one-time setup. It is a **lifecycle**: define what must be protected, prevent what you can, detect what still happens, respond under pressure, recover safely, and evaluate whether the controls actually work.

Hospital example

St. Isidore does not simply install a firewall and stop. It updates policies, trains staff, monitors alerts, practices ransomware recovery, tests backups, and reviews incidents to improve the next cycle.

Scheme: Security Lifecycle



Reading the scheme

Each incident, audit, and recovery test should feed back into better policy and prevention.

Prevention

Concept

Prevention aims to stop attacks before they succeed.

Hospital example

Patch management, MFA, network segmentation, secure configuration, and phishing-resistant authentication reduce the chance of compromise.

Detection

Concept

Detection identifies suspicious activity, misuse, policy violations, or confirmed intrusions.

Hospital example

An alert triggers when one account opens hundreds of patient records across unrelated wards in ten minutes.

Response

Concept

Response contains an incident, limits further damage, preserves evidence, and coordinates communication.

Hospital example

When malware is detected, IT isolates the workstation, disables the account, informs clinical leadership, and checks whether patient care is affected.

Concept

Recovery restores services and data to a trustworthy state after an incident.

Hospital example

After ransomware, the hospital rebuilds servers from clean images, restores tested backups, and reconciles paper medication orders entered during downtime.

Concept

Assurance is confidence that security measures satisfy their requirements.

Hospital example

The hospital does not merely assume backups work. It performs restoration tests and records the time needed to recover critical systems.

Evaluation

Concept

Evaluation checks a system against defined criteria through testing, review, audit, certification, or formal analysis.

Hospital example

Before buying a new telemedicine platform, St. Isidore reviews encryption, logging, access control, data retention, vendor support, and incident notification terms.

Recap: Design and Policy

Takeaways

- Design principles reduce complexity and limit damage when something fails.
- Policy translates security goals into organisational rules and responsibilities.
- Security is a trade-off among protection, usability, cost, performance, and care delivery.
- Prevention, detection, response, recovery, assurance, and evaluation form a lifecycle.

Hospital checkpoint

An emergency-access feature is acceptable only if it is fast enough for care, limited in scope, logged, reviewed, and tested.

Hash Functions

Section Context: Hash Functions

What we are talking about

Before we encrypt or authenticate messages, we need the idea of a digest: a compact fingerprint that changes when data changes.

Hospital lens

When a software update, discharge summary, or configuration file arrives, the hospital needs a way to detect whether the exact bytes changed.

Hash Functions

Concept

A **hash function** maps data of variable length to a fixed-length digest. A secure hash acts like a fingerprint of the input.

Hospital example

The hospital computes a hash of a software update before installation. If the downloaded file has changed, the hash will not match the expected value.

Properties of Secure Hashes

- **Preimage resistance:** given a digest, it is infeasible to find a matching input.
- **Second-preimage resistance:** given one input, it is infeasible to find a different input with the same digest.
- **Collision resistance:** it is infeasible to find any two different inputs with the same digest.

Hospital example

These properties help protect password storage, file-integrity monitoring, and digital signatures.

Hashing Passwords

Concept

Systems should store password verifiers, not plaintext passwords. In practice this requires password-hashing schemes with salts and work factors, not a bare fast hash.

Hospital example

If an attacker steals the staff account database, salted password hashes make password recovery harder than if plaintext passwords were stored.

File Integrity Monitoring

Concept

Hash values can help detect whether important files changed. The reference hashes must be protected from attackers.

Hospital example

St. Isidore stores trusted hashes of critical EHR configuration files. A nightly check alerts IT if malware modifies them.

Standard family

The Secure Hash Standard defines SHA algorithms used in many security applications [National Institute of Standards and Technology, 2015]. Algorithm choice matters because older hashes may become weak over time.

Hospital example

The hospital prefers modern SHA-2 or SHA-3 based mechanisms where appropriate, and avoids legacy uses of broken hashes in certificates or signatures.

Recap: Hash Functions

Takeaways

- Hash functions produce fixed-length fingerprints of data.
- They support file checks, password storage, MACs, and digital signatures.
- Hashes do not hide data: they are not encryption.

Encryption Basics

Section Context: Cryptographic Foundations

What we are talking about

Cryptography transforms information so that only authorised parties can read it, verify it, or trust where it came from. We start with the basic ideas before looking at specific families of algorithms.

Hospital lens

At St. Isidore, cryptography protects a lab result while it travels, a backup while it is stored, and a prescription order while the pharmacy verifies who sent it.

Kerckhoffs's Principle

Principle

The security of a cryptosystem should depend on the secrecy of the **key**, not on the secrecy of the **algorithm** [Kerckhoffs, 1883].

Hospital example

St. Isidore should not rely on a vendor's secret home-made encryption method. It should rely on well-studied algorithms and protect the hospital's keys carefully.

What Cryptography Provides

Core services

Cryptography can support **confidentiality**, **authentication**, **integrity**, and in some systems **anonymity**.

- **Confidentiality**: hiding content from unauthorised readers.
- **Authentication**: verifying identities or sources.
- **Integrity**: detecting unauthorised modification.
- **Anonymity**: reducing linkability to a person or communication path.

Healthcare Example: One Lab Result

Hospital example

A potassium result moves from the laboratory system to the EHR. Cryptography can keep the value private, prove it came from the lab, detect changes, and help prevent replay of an old result.

Takeaway

Encryption is not only about secrecy. Clinical systems also need authenticity, integrity, freshness, and correct ordering.

Encryption and Decryption

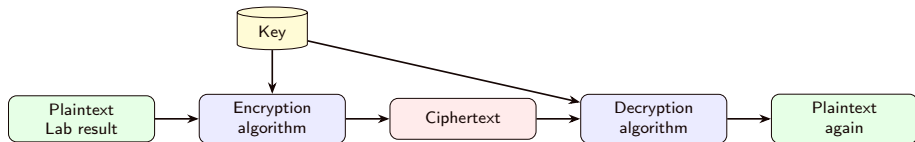
Concept

Encryption transforms plaintext into ciphertext. **Decryption** transforms ciphertext back into plaintext using the correct key.

Hospital example

Before sending a discharge letter to a referral clinic, the hospital encrypts it. If the message is intercepted, the attacker sees ciphertext rather than patient data.

Scheme: Encryption Flow



Reading the scheme

The algorithm may be public. The key is the secret that controls who can decrypt.

Classical Ciphers

Concept

Classical ciphers such as the Caesar cipher replace or move symbols according to a simple rule. They are useful for teaching, but not for real security.

Hospital example

Changing every letter in a patient's name by three positions would not protect the record. Anyone could reverse it once the pattern is recognised.

Why Simple Ciphers Fail

Concept

Substitution and transposition ciphers preserve patterns. Attackers can use language statistics, known formats, and repeated phrases to recover meaning.

Hospital example

Hospital messages often repeat phrases such as “patient ID”, “diagnosis”, or “lab result”. Repeated structure gives attackers clues if the cipher leaks patterns.

Recap: Cryptographic Foundations

Takeaways

- Cryptographic security should depend on protected keys, not secret algorithms.
- Encryption turns plaintext into ciphertext; decryption reverses it with the right key.
- Healthcare systems need secrecy, authenticity, integrity, and freshness.
- Simple historical ciphers fail because they leak patterns.

Symmetric Encryption

Section Context: Symmetric Encryption

What we are talking about

Symmetric encryption uses the **same secret key** for encryption and decryption. It is fast, widely deployed, and central to protecting stored and transmitted healthcare data.

Hospital lens

When St. Isidore encrypts database backups or file transfers, it commonly uses symmetric cryptography under the hood.

Symmetric Encryption

Concept

In symmetric encryption, sender and receiver share one secret key. Anyone with the key can decrypt the protected data.

Hospital example

The backup system encrypts nightly EHR backups with a hospital-controlled key. Recovery is possible only if that key remains available and protected.

Components of a Symmetric System

- **Plaintext:** original data.
- **Encryption algorithm:** transforms the plaintext.
- **Secret key:** controls the transformation.
- **Ciphertext:** encrypted output.
- **Decryption algorithm:** reverses the process with the same key.

Hospital example

The plaintext is an EHR export; the key is stored in a key-management system; the ciphertext is the encrypted backup file.

Two Requirements for Security

Concept

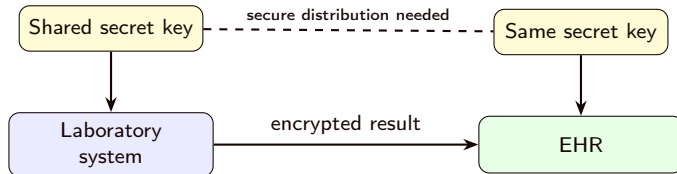
A symmetric system needs both a **strong algorithm** and **secure key handling**.

- The algorithm should resist cryptanalytic and brute-force attacks.
- The key must be generated, distributed, stored, rotated, and retired securely.

Hospital example

AES is not enough if the encryption key is stored in a shared spreadsheet or left on the same server as the data it protects.

Scheme: Symmetric Key Sharing



Takeaway

The difficult part is often not encryption itself, but getting the key safely to the right systems.

Cryptanalytic Attacks

Concept

Cryptanalysis tries to exploit weaknesses in the algorithm, message structure, or available plaintext/ciphertext examples.

Hospital example

If every exported report begins with the same header, a weak encryption mode may reveal patterns that help an attacker reason about the protected files.

Brute-Force Attacks

Concept

Brute force tries possible keys until the output looks like meaningful plaintext. Longer keys make exhaustive search infeasible.

Hospital example

If archived patient files were encrypted with an obsolete short key, an attacker with copied ciphertext may be able to recover the records later.

Modern standard

The Advanced Encryption Standard is a symmetric block cipher with 128-bit blocks and key sizes of 128, 192, or 256 bits [National Institute of Standards and Technology, 2023a].

Hospital example

AES is commonly used to encrypt backups, databases, disks, VPN traffic, and secure application channels in healthcare environments.

Recap: Symmetric Encryption

Takeaways

- Symmetric encryption is fast and widely used.
- Security depends on strong algorithms and strong key management.
- AES is the modern standard example to recognise in this recap.
- A block cipher still needs a safe mode when used on real messages.
- Key exposure compromises every message protected with that key.

Diffie-Hellman Key Agreement

Diffie-Hellman Key Agreement

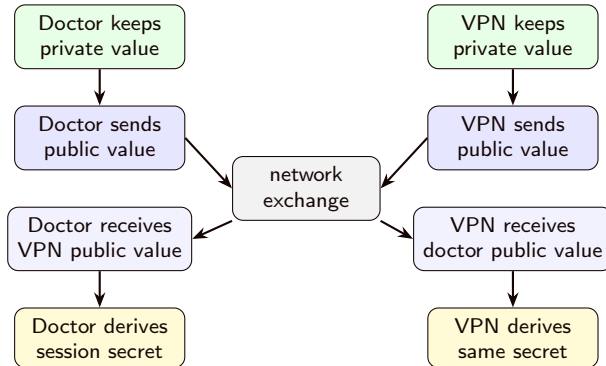
Concept

Diffie-Hellman allows two parties to establish a shared secret over an insecure channel [Diffie and Hellman, 1976]. By itself, basic Diffie-Hellman does not authenticate who is on the other side.

Hospital example

A doctor connects to the VPN from home. The protocol can agree on session keys, but certificates and authentication are needed to prevent an attacker from impersonating the hospital.

Scheme: Diffie-Hellman



Reading the scheme

Only public values cross the network. Each side keeps its private value local and combines it with the received public value, deriving the same session secret independently.

Recap: Diffie-Hellman

Takeaways

- Diffie-Hellman is key agreement, not document encryption.
- The shared secret is derived locally rather than transmitted.
- Authentication is still needed to avoid impersonation.

Message Authentication Codes

Message Authentication

Concept

A message is authentic when the receiver can verify its source and detect changes to its content. Good authentication can also check sequence and freshness.

Hospital example

The EHR should reject a lab result if it was modified, replayed from yesterday, or sent by a system pretending to be the laboratory.

Why Encryption Alone Is Not Enough

Concept

Confidentiality does not automatically imply integrity. Some encrypted data can still be reordered, replayed, or tampered with unless authentication is included.

Hospital example

If encrypted medication-order blocks can be rearranged without detection, the receiver may decrypt valid-looking but clinically wrong content.

Message Authentication Code

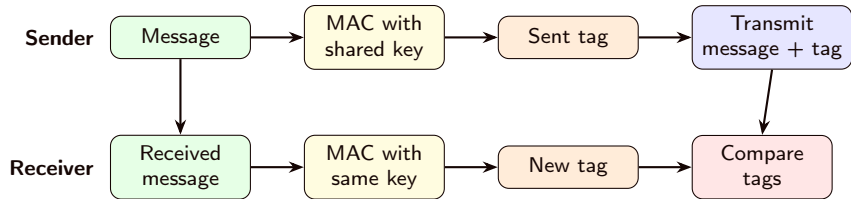
Concept

A **Message Authentication Code** (MAC) is a short tag computed from a message and a shared secret key. The receiver recomputes the tag and compares it.

Hospital example

The lab system attaches a MAC to each result message. The EHR verifies the MAC before accepting the result into the patient record.

Scheme: MAC Verification



Takeaway

The receiver does not decrypt a MAC. It recomputes the tag from the received message and the shared secret, then compares the two tags.

Recap: Message Authentication Codes

Takeaways

- MACs use a shared secret key, so they naturally follow symmetric-key ideas.
- They detect tampering and authenticate messages between systems that share the key.
- They complement encryption: confidentiality alone does not prove integrity.

Asymmetric Encryption with RSA

Section Context: Asymmetric Encryption

What we are talking about

Asymmetric encryption uses a public key and a private key to solve a specific problem: sending protected data or a protected session key without first sharing a secret.

Hospital lens

St. Isidore can publish a receiving service public key. External clinics use it to protect referral packages or the short session keys that protect larger files.

Asymmetric Keys

Concept

Each participant has a **public key** that can be shared and a **private key** that must stay secret. What one key does, the paired key can verify or reverse, depending on the scheme.

Hospital example

The patient portal publishes a certificate containing its public key. The matching private key stays protected on the server or in a hardware security module.

Confidentiality with Public Keys

Concept

To send confidential data to Alice, Bob encrypts using Alice's public key. Only Alice's private key can decrypt.

Hospital example

An external clinic can encrypt a referral package so that only St. Isidore's receiving service can open it.

Families of Public-Key Algorithms

What changes between algorithms

Public-key algorithms are not interchangeable names. Each family is built around a hard mathematical problem and is usually designed for one main purpose: **encryption**, **digital signature**, or **key agreement**.

Hospital example

The patient portal certificate may use one algorithm to sign its identity, while the VPN handshake uses another algorithm to agree on fresh session keys. Both are public-key cryptography, but they do different jobs.

Concept

RSA is based on modular arithmetic with a public modulus. The public key can transform data in one direction, while the private key provides the **trapdoor** needed to reverse or sign that transformation.

Hospital example

An external clinic can encrypt a small session key with St. Isidore's RSA public key. Only the hospital private key can unwrap it. The same RSA key family can also sign messages when used with a signature padding scheme.

RSA in Practice

Important distinction

RSA should not be used to encrypt large clinical files directly. It is usually used to protect small values, such as a symmetric session key, or to create digital signatures.

Hospital example

For an imaging report, St. Isidore encrypts the file with AES, then uses RSA only to protect the AES key. This is the digital-envelope pattern used later in the lesson.

Recap: Asymmetric Encryption

Takeaways

- Public-key cryptography uses key pairs, not one shared secret.
- Public keys can be shared; private keys must be protected.
- RSA is the main public-key encryption example in this recap.
- It usually works together with symmetric encryption in real systems.

Digital Signatures

Section Context: Digital Signatures

What we are talking about

Digital signatures use public-key cryptography to verify origin and integrity. The focus here is on what a signature proves, what it does not prove, and how verification works.

Hospital lens

The hospital must verify that vendor updates, formulary changes, and signed clinical documents really came from the expected source and were not modified.

Digital Signatures

Concept

A digital signature is usually created by hashing a message and signing the hash with the sender's private key.

Hospital example

The pharmacy service signs a formulary update. The EHR verifies the signature before using the new medication rules.

DSS Signature Algorithms

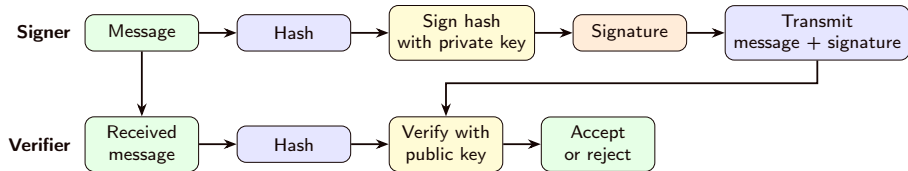
Algorithm idea

The Digital Signature Standard (DSS) specifies approved digital-signature algorithms [National Institute of Standards and Technology, 2023b]. Current FIPS 186-5 covers **RSA signatures**, **ECDSA**, and **EdDSA**. These are signature mechanisms: they do **not** encrypt data and they do **not** establish shared keys.

Hospital example

A software vendor can sign a pacemaker gateway firmware update. The hospital verifies the signature before installation to check authenticity and integrity, but the signature itself does not hide the update contents.

Scheme: Digital Signature



Takeaway

The signature protects the hash of the message. Verification uses the sender's public key and the freshly computed hash of the received message.

Recap: Digital Signatures

Takeaways

- Digital signatures authenticate origin and detect modification.
- The signer uses a private key; verifiers use the matching public key.
- Signatures do not hide content; combine them with encryption when confidentiality is needed.

Digital Envelopes

Section Context: Digital Envelopes

What we are talking about

Real systems rarely use only one cryptographic primitive. A digital envelope combines fast symmetric encryption with public-key protection of the temporary key.

Hospital lens

A large imaging report can be encrypted efficiently with a fresh session key, while the session key is protected for the intended specialist using that specialist's public key.

Digital Envelopes

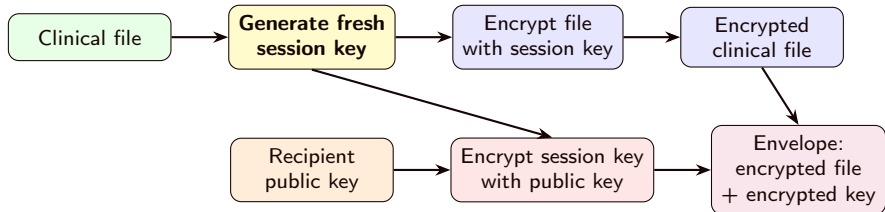
Concept

A **digital envelope** combines symmetric and public-key cryptography: encrypt the message with a fresh symmetric key, then encrypt that key with the recipient's public key.

Hospital example

The hospital sends a large imaging report to an external specialist. The file is encrypted with a fast session key; the session key is protected with the specialist's certified public key.

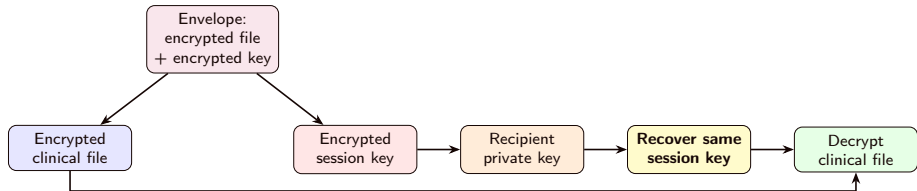
Scheme: Digital Envelope: Sender



Reading the scheme

The sender creates a fresh session key. The key encrypts the file, then the recipient's public key encrypts that session key.

Scheme: Digital Envelope: Receiver



Reading the scheme

The receiver uses the private key only to recover the session key. The recovered session key then decrypts the clinical file.

Cryptographic Roles: Recap Map

Technique	Main role	Hospital example
AES	Bulk encryption	Backups and clinical files
RSA	Key wrapping / small encryption	Protecting a session key
Diffie-Hellman	Key agreement	Fresh VPN or TLS keys
Hash functions	Integrity fingerprint	Checking software updates
MAC	Shared-secret authentication	Verifying lab-result messages
RSA signatures	Signature algorithms	Verifying signed updates or documents

Important

Complex systems combine these roles: public-key methods establish trust or protect keys, symmetric encryption protects bulk data, and hashes, MACs, or signatures detect tampering.

Additional Materials

Additional Practice References

Caesar cipher playground

For a quick visual experiment with a historical substitution cipher, use Cryptii's Caesar cipher tool:

<https://cryptii.com/pipes/caesar-cipher/>

How to use it

Encrypt a short hospital-style phrase such as LAB RESULT READY, then change the shift and observe how easy the structure remains to test by hand.

Additional Practice Tools: Encryption

RSA tool

For a quick experiment with public-key encryption, use this RSA tool:

https:

[//www.devglan.com/online-tools/rsa-encryption-decryption](https://www.devglan.com/online-tools/rsa-encryption-decryption)

How to use it

Try encrypting a short referral note with a public key, then observe that the matching private key is needed to recover the original text.

Additional Practice Tools: Symmetric Encryption



AES tool

For a quick experiment with symmetric encryption, use this AES tool:
<https://www.devglan.com/online-tools/aes-encryption-decryption>

How to use it

Encrypt a short message such as WARD BACKUP READY and verify that changing the key prevents correct decryption.

Additional Practice Tools: Hashes and HMAC

SHA file hash

For a quick experiment with file fingerprints, use this SHA file hash generator:

<https://www.devglan.com/online-tools/file-hash-generator>

How to use it

Hash a small text file, change one character, and compare the two outputs to see how sensitive a hash is to modification.

Additional Practice Tools: HMAC

HMAC-SHA256 tool

For a quick experiment with keyed message authentication, use this HMAC-SHA256 tool:

<https://www.devglan.com/online-tools/hmac-sha256-online>

How to use it

Compute a tag for a lab-result message, then change either the message or the shared secret and observe that the tag changes.

Additional Reading on Malware

Malware taxonomy

Fortinet's malware overview is useful as a glossary-style reference for common malware categories:

https:

[//www.fortinet.com/resources/cyberglossary/types-of-malware](https://www.fortinet.com/resources/cyberglossary/types-of-malware)

How to use it

Use it to connect Lesson 1 threat vocabulary with operational examples: ransomware, spyware, worms, trojans, and social-engineering delivery paths.

Healthcare Data Privacy and Security

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